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Research on the construction and application of retrieval enhanced generation (RAG) model based on knowledge graph

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Generative pre-trained language models have demonstrated strong capabilities in natural language processing tasks, but they still suffer from "fact hallucination" and weak knowledge timeliness in open-domain question answering and text generation. To improve the accuracy and knowledge consistency of generated content, this paper proposes a Knowledge Graph-based Retrieval Enhanced Generation Model (KG-RAG), which integrates structured knowledge graphs into traditional RAG architectures to enhance the model's ability to understand semantic and inferential relationships. The model designs a dual-channel retrieval mechanism: on one hand, it uses Dense Passage Retrieval (DPR) for vectorized retrieval of unstructured texts; on the other hand, it employs graph neural networks (GNN) to structurally retrieve semantic paths within the knowledge graph, and through path attention mechanisms, it filters out the most relevant entity relationship chains to guide the knowledge injection module. Experimental results on the Natural Questions and PubMedQA datasets show that KG-RAG outperforms the original RAG model across multiple evaluation metrics. On the Natural Questions dataset, the ROUGE-L score of the KG-RAG model improves from 41.2 to 46.9, the BLEU score rises from 31.5 to 38.7, and the FactScore increases by 13.6%, significantly enhancing the knowledge consistency of the generated text. In the PubMedQA task, KG-RAG achieves an accuracy rate of 81% in medical question answering. 3%, an improvement of 6.8% points over RAG, demonstrates its advantage in knowledge reasoning within specialized fields. Furthermore, case studies show that KG-RAG can effectively integrate entity paths from knowledge graphs to generate more logical and factual answers in complex question-answering tasks. This method has broad application prospects in intelligent question-answering systems, multi-turn conversations, and educational Q&A scenarios. Future research will consider introducing dynamic knowledge update mechanisms and multimodal graph information to further enhance the capabilities and adaptability of KG-RAG in real-world tasks.

Keywords Retrieval enhanced generation, Knowledge graph, Natural language generation, Question answering system (Question answering)

In recent years, with the widespread application of pre-trained language models (such as BERT, GPT, etc.) in natural language processing (NLP), generative models have achieved significant results in tasks like text generation, question answering systems, and dialogue systems. However, traditional generative models generally rely on fixed parameters to store knowledge internally, leading to issues such as "knowledge staleness" and "fact hallucination" when handling open-domain questions or factual text generation, which affect the accuracy and credibility of the generated results[1].

To address the limitations in knowledge acquisition for generative models, Retrieval-Augmented Generation (Retrieval-Augmented Generation, RAG) models have been proposed. Representative methods include RAG (Lewis et al., 2020), which combines a document retriever with a generator to enable the model to dynamically access external knowledge before generating text, thereby enhancing the timeliness and accuracy of knowledge. However, most existing RAG methods rely on unstructured text repositories (such as Wikipedia), which have limitations in semantic expression, logical reasoning, and entity relationship modeling. Meanwhile, Knowledge Graphs (Knowledge Graph, KG), as structured knowledge carriers, excel in entity relationship representation

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and can provide interpretable semantic paths and inference clues, showing great potential in question answering and generation tasks[2].

To this end, this paper proposes a knowledge graph-enhanced retrieval generation model (KG-RAG). By constructing graph neural networks and path attention mechanisms, it effectively models and utilizes relevant entities and relationships in the graph. It also combines the results of unstructured document retrieval to jointly guide text generation. We conducted empirical studies on datasets such as Natural Questions and PubMedQA, verifying the model's advantages in text accuracy, fact consistency, and professional domain question answering capabilities[3].

The main contributions of this paper are as follows: (1) A KG-RAG architecture that integrates graph structure and semantic path information is constructed; (2) A generation fusion mechanism that combines structured and unstructured retrieval results is designed; (3) The performance of KG-RAG is systematically evaluated across multiple tasks, validating its broad applicability in both general and specialized domain tasks.

Research on technical theory

Retrieval-Augmented Generation (RAG) has emerged as a powerful framework to enhance the factual consistency and timeliness of large language models (LLMs) by retrieving external knowledge. Traditional RAG methods primarily rely on unstructured text corpora, which limits their ability to handle complex relationships and perform multi-hop reasoning. To address this, recent research has explored the integration of structured knowledge graphs (KGs) into RAG frameworks, giving rise to KG-RAG models.

Sanmartin[4] (2024) introduced a model named KG-RAG that incorporates structured knowledge into the RAG pipeline to reduce hallucinations and improve answer consistency. However, the model lacks fine-grained semantic modeling of entities and struggles with multi-hop reasoning tasks.

Mavromatis and Karypis (2024)[5] proposed GNN-RAG, which leverages Graph Neural Networks (GNNs) for retrieval, allowing better representation of complex entity relations. While the model improves reasoning, it faces scalability and computational overhead challenges.

Xu et al. (2024)[6] proposed a dual-pathway KG-RAG that retrieves information through both entity paths and contextual evidence. Their method shows strong performance on open-domain and medical QA tasks by capturing complementary signals from structured and unstructured sources.

Su et al. (2024)[7] presented a knowledge graph-based agent tailored for complex medical question answering, focusing on dynamic updates and interpretability. This demonstrates how domain-specific KG-RAG systems can provide high-accuracy responses in sensitive applications.

In addition, Amugongo et al. (2024)[8] conducted a systematic review of RAG methods in healthcare, highlighting three major challenges in KG-RAG models: (1) weak integration between structured and unstructured knowledge, (2) limited accuracy in selecting multi-hop reasoning paths, and (3) lack of standard evaluation protocols.

In comparison, our proposed KG-RAG model improves upon previous works in three key aspects:

Path attention mechanism that enhances the accuracy of selecting semantically relevant reasoning paths;

Dual-channel retrieval architecture that combines KG-based structure with contextual semantic retrieval to improve recall and coverage;

Prompt fusion strategy that enables more effective integration of retrieved structured and unstructured information into the generation pipeline.

These contributions are elaborated in the methodology and further validated through experiments presented in the following sections.

Evolution of the RAG model

Generative models have made significant progress in natural language processing tasks, but they face issues such as untimely knowledge access and low generation accuracy when handling open-domain question answering or long text generation. To address these challenges, Facebook proposed the Retrieval-Enhanced Generation (Retrieval-Augmented Generation, RAG) architecture, which combines a retriever (Retriever) and a generator (Generator). By retrieving relevant knowledge and integrating it with the output of the generation model, this approach effectively enhances the performance of generation tasks[9].

The core idea of the RAG model is to integrate generative models with external knowledge sources (such as document libraries), enabling the generator to produce text with higher semantic accuracy based on retrieved relevant information. Unlike traditional generative models (like GPT and BART) that rely solely on internal parameter libraries, the RAG model dynamically accesses knowledge bases through a retrieval module and combines the retrieved information with the input context via the generator to produce reasonable outputs[10].

1. The structure and principle of RAG architecture are shown in Fig. 1:

The architecture of the RAG model is mainly composed of two parts: the retriever (Retriever) and the generator (Generator)[11].

Retrieval Engine (Retriever): The RAG model first retrieves documents related to the input query through a retrieval module. To improve retrieval efficiency, RAG employs dense retrieval (Dense Retrieval) techniques, such as DPR (Dense Passage Retrieval). This method converts text and queries into vector q representations and uses similarity d calculations to retrieve the most relevant documents from the document repository. Specifically, given a query, the retrieval engine selects relevant documents by calculating the similarity scores between the query and the documents:

$$score(q, d) = q^T d \tag{1}$$

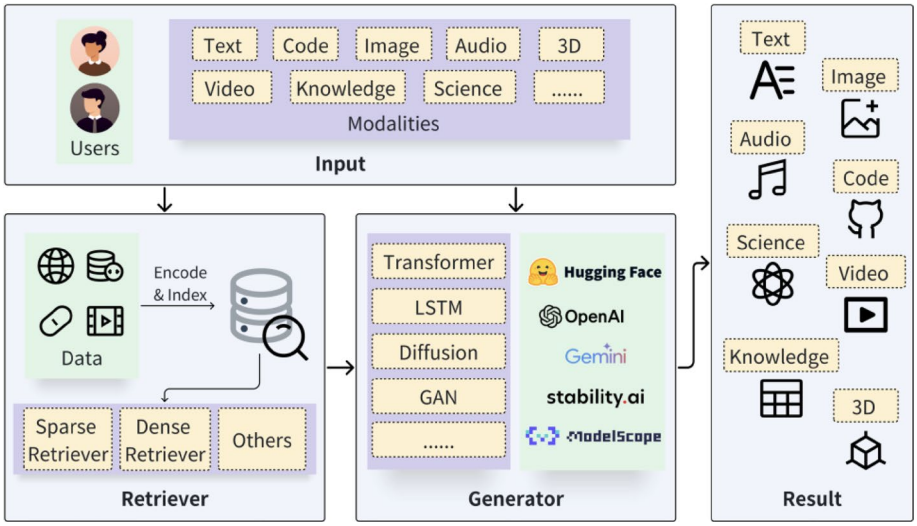

Fig. 1. RAG architecture.

Among them qd , and are the vector representations T of query and document, respectively, representing the transposition operation.

Generator (Generator): After retrieving relevant documents, the RAG model uses a generator (usually BART or T5) to integrate the retrieved documents with the original query, generating the final answer or text. The generator takes the document and query as input and produces a coherent and context-appropriate text output[12]. Its generation process is based on the following conditional probabilities:

$$P(y|x, D) = \prod_{t=1}^{T} P(y_t|y < t, x, D) \tag{2}$$

Among them y_t, the generated x word is the D input query, the $y < t$ retrieved relevant document is the previously generated word.

The advantage of the RAG model is that the generator no longer relies solely on internal knowledge of the model, but combines retrieved external knowledge, addressing the "illusion of fact" problem faced by traditional generative models. At the same time, the RAG model can effectively utilize knowledge contained in large-scale document libraries, enhancing the richness and accuracy of generated content[13].

However, the RAG model still faces some challenges. First, the performance of the retrieval module significantly impacts the quality of the final output; the relevance and diversity of the retrieved results directly affect the quality of the generated content. Second, when integrating various retrieval information, the generator needs to effectively select and merge multiple candidate documents retrieved, to avoid information conflicts and inconsistencies during the generation process[14].

To address the defects of traditional Retrieval-Augmented Generation (RAG) in retrieval relevance and diversity, the Knowledge Graph-enhanced Retrieval-Augmented Generation (KG-RAG) model proposed in this study introduces a structured knowledge graph as an intermediate retrieval layer and constructs a "text + knowledge graph" dual-channel retrieval architecture. Specifically, the domain-adapted fine-tuning of Dense Passage Retrieval (DPR) in the text channel is employed to improve the accuracy of semantic matching; the path attention scoring mechanism in the knowledge graph channel is used to filter out highly relevant entity relationship chains; meanwhile, diversity constraint strategies are supplemented (i.e., penalizing redundant paragraphs in the text channel and controlling the coverage of relationship types in the knowledge graph channel). These measures optimize retrieval quality at the source and provide "highly relevant + multi-dimensional" knowledge support for the subsequent generation module—this design also constitutes one of the core innovations that distinguish KG-RAG from traditional single-channel RAG.

It is precisely the inherent flaws of traditional RAG in retrieval relevance (semantic deviation) and diversity (homogeneity) that serve as the core motivation for the design of the "text + knowledge graph" dual-channel architecture of KG-RAG in this study. By means of dual-channel complementarity and targeted optimization, the bottleneck of generation quality is addressed at the source of retrieval.

Application of knowledge graph in NLP

In natural language processing (NLP) tasks, traditional generation models and understanding models typically rely on extensive training with text data. However, with the introduction of knowledge graphs (KGs) as structured knowledge resources, researchers have begun to explore how to leverage the entities, relationships, and reasoning capabilities of KGs to enhance the performance of language models. As shown in Fig. 2, knowledge graph-based reasoning models enhance the model's reasoning and semantic understanding capabilities by integrating

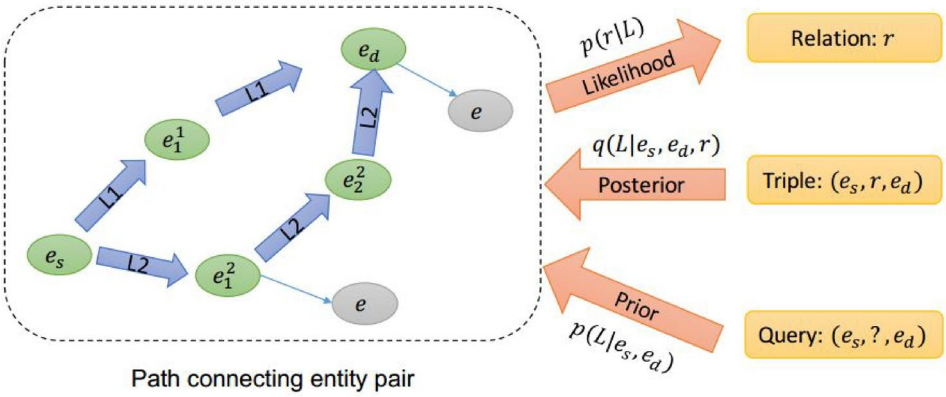

Fig. 2. Reasoning model based on KG.

knowledge from graph structures providing strong support for many tasks such as question answering, dialogue generation, and information extraction[15].

(1) K-BERT model

K-BERT (Knowledge-enhanced BERT) is an extension of BERT (Bidirectional Encoder Representations from Transformers), aiming to embed knowledge graph information into the BERT model, thereby enhancing BERT's performance in tasks involving knowledge reasoning. The core idea of K-BERT is to introduce entity information from the KG by adding entity representations to BERT's input, helping the model understand more complex semantic relationships[16].

Principle: K-BERT enhances BERT's contextual modeling capability by inserting entity representations from KG into the input text. Specifically, K-BERT replaces entities in the text with corresponding entity vectors and inputs them along with the original word embeddings into the BERT model. Entity vectors are typically obtained as[17] from the embedding representations of entities in the knowledge graph (such as TransE, DistMult, etc.).

Assuming the input $T = \{\omega_1, \omega_2, \omega_3 \ldots \omega_n\}$ text is ω_i , where i is the word ω_i at the-th position, if is an entity, K-BERT replaces it with the e_i KG embedding representation of the corresponding entity. $X = \{x_1, x_2, x_3 \ldots x_n\}$ Then, the model's input will be a sequence of vectors that mix word embeddings and entity embeddings, where:

$$x_i = \begin{cases} embedding(\omega_i) \\ KG_embedding(e_i) \end{cases} \tag{3}$$

The output of K-BERT is generated by the Transformer layer of BERT to generate context-related representations for subsequent tasks.

Formula: The optimization objective of K-BERT depends on the standard training objective of BERT model, namely, maximizing the likelihood function [18] of masked language X model (Masked Language Model, MLM). For a given input sequence, the objective of the model is to maximize the following formula:

$$L_{MLM} = -\sum_{i=1}^{n} \log P(\omega_i | X_{-i}) \tag{4}$$

Among them X_{-i} , the context i input without the ω_i word i is the real label of the word.

(2) Knowledge graph-based reasoning models can be applied to multiple NLP tasks, especially in question answering, dialogue generation, and text inference. For example, the model can infer missing information through the relationship chains of the knowledge graph or generate more logically consistent responses in conversations. Reasoning tasks based on KGs typically include entity linking, relation reasoning, event inference, etc.[19]

Entity Linking and Reasoning: The entity linking task aims to match mentions in text with entities in a knowledge graph, while the reasoning task leverages relationships and inference rules within the graph to derive new knowledge. For example, in a question answering system, if the input question is "Who was Einstein's mentor?", the model can use entity linking to identify "Einstein" as an entity, and then reason through the knowledge graph to derive information about his mentors[20].

The current situation and problems of fusion between 2.3 KG and generation model

In recent years, with the rise of pre-trained language models (such as BERT, T5, and the GPT series), knowledge graphs (KGs), as important carriers of structured semantic information, have been widely applied to enhance

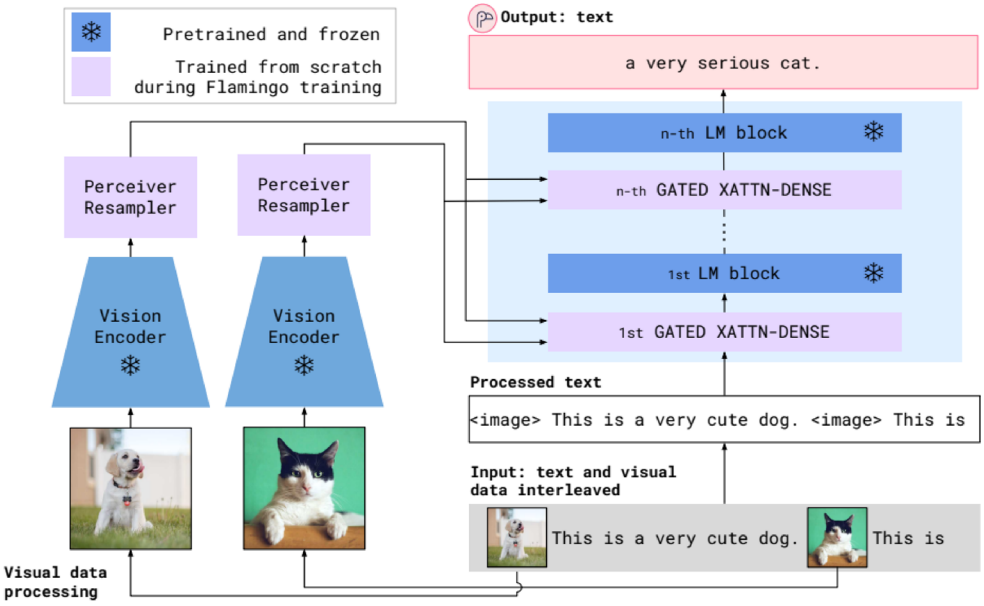

Fig. 3. Generative model.

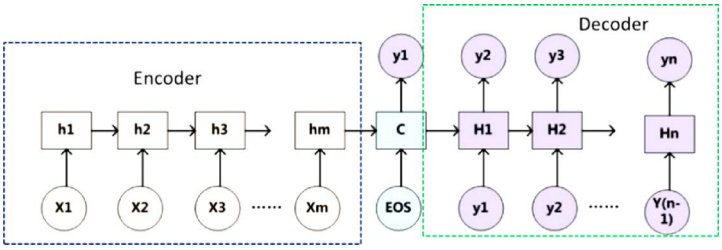

Fig. 4. Design of Encoder-Decoder structure.

the language understanding and reasoning capabilities of generative models, as shown in Fig. 3. To efficiently integrate entity and relationship information from KGs into generative tasks, researchers have proposed various fusion strategies[21].

KG Embedding Embedding into encoder (Encoder) In this strategy, entities and relationships in the knowledge graph are first embedded (such as TransE, RotatE, ComplEx, etc.), and then these embedded vectors are concatenated or fused with the word embedding of the text, and sent to the encoder for semantic modeling together.

The Encoder-Decoder structure is a universal framework composed of two recurrent neural networks (RNNs). The RNN architecture can be replaced with Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) to address the issue of information loss during transmission. The Encoder and Decoder parts can be any text, speech, image, or video data; the model can be CNN, RNN, LSTM, GRU, Attention, or other[22]. Encoding involves converting the input sequence into a fixed-length vector, while decoding is the process of transforming the previously generated fixed vector back into an output sequence, as shown in Fig. 4.

In addition, the introduction of the Attention mechanism Source is composed of a series of components. Given an element Query in Target, by calculating the similarity between Query and each Key, the weight coefficient of each Key for Value is obtained. Then, the weighted sum of Values is calculated to get the final Attention value. Introducing Multi-Head Attention divides the model into multiple heads, forming several subspaces, allowing the model to focus on information from different directions Transformer or specific layers of Bert have unique functions, with the lower layers focusing more on syntax and the higher layers focusing more on semantics[23], as shown in Fig. 5.

Recent studies have shown that while traditional RAG models effectively enhance the factuality and timeliness of language generation, they struggle with tasks that require structured reasoning over complex entity relationships. Specifically, several challenges persist:

First, text-based retrievers lack the ability to explicitly capture relational semantics encoded in knowledge graphs, which are essential for understanding context-dependent or hierarchical entity links.

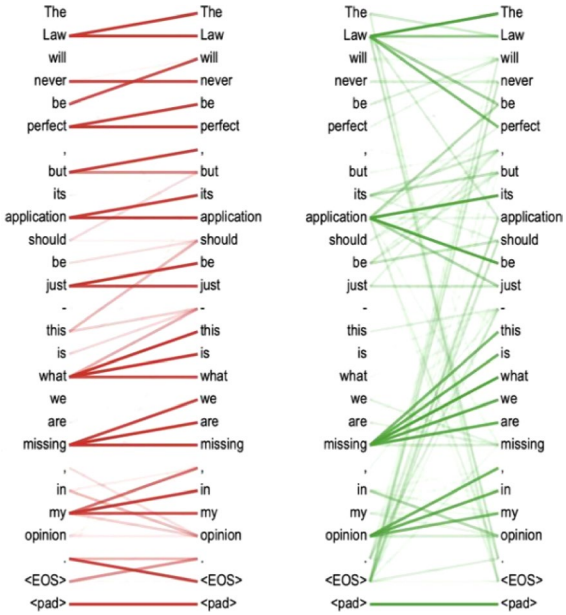

Fig. 5. Structure design of multi-head self-attention mechanism.

Second, multi-hop reasoning remains difficult due to the existence of numerous redundant or semantically weak paths in the knowledge graph. Selecting relevant reasoning chains requires not only accurate path ranking but also semantic filtering to avoid drift and contradiction.

Third, the integration of structured and unstructured knowledge during generation is often superficial, as most existing approaches concatenate retrieved results or rely on fixed templates, leading to sub-optimal reasoning performance.

Lastly, current RAG systems lack dynamic adaptability to evolving knowledge, especially in domains like medicine where facts change frequently and rely heavily on up-to-date structured sources.

These limitations motivate the design of our KG-RAG model, which introduces (1) a path attention mechanism to improve multi-hop path selection; (2) dual-channel retrieval to integrate both structured graph data and unstructured text; and (3) a prompt fusion strategy to enhance semantic alignment between retrieved knowledge and generation inputs.

Method design

Overall architecture design: KG-RAG model

This study proposes a retrieval-enhanced generation model (KG-RAG, Knowledge Graph-enhanced Retrieval-Augmented Generation) that integrates structured knowledge graphs. The aim is to improve the semantic understanding, entity consistency, and generation accuracy of traditional RAG models in complex question-answering tasks. The core innovation of KG-RAG lies in introducing structured knowledge graphs as an intermediate retrieval layer, replacing or enhancing traditional document vector libraries, thereby achieving more precise entity-level retrieval and inference[24,25].

The KG-RAG is composed of three core modules: knowledge graph module (KG Module), retrieval generation module (RAG Core) and control mechanism module (Path Controller), as shown in Fig. 6.

Knowledge graph module

The Knowledge Graph module is the foundation of the KG-RAG architecture, primarily responsible for constructing, storing, and querying entities and their semantic relationships, supporting subsequent semantic retrieval and multi-hop path inference. This module is built on Neo4j graph database and supplemented with custom domain ontologies to enhance semantic consistency and structured expression capabilities.

(1) The construction process of knowledge graph.

The construction of KG is divided into the following key steps:

Entity recognition (NER): The Fine-tuned BERT-BiLSTM-CRF model is used to identify proper nouns, institutions, places, terms and other entities in the original corpus.

Relation extraction (RE): Using BERT + Multi-Head Attention relation classifier, syntactic and semantic connections between entities are classified to form candidate relation pairs.

Entity Disambiguation and Linking (EL): Uniquely identify ambiguous entities by combining context and external knowledge bases (such as Wikidata, CN-DBpedia).

Triple generation: The above processed entities and relationships are combined to generate standard triple form:

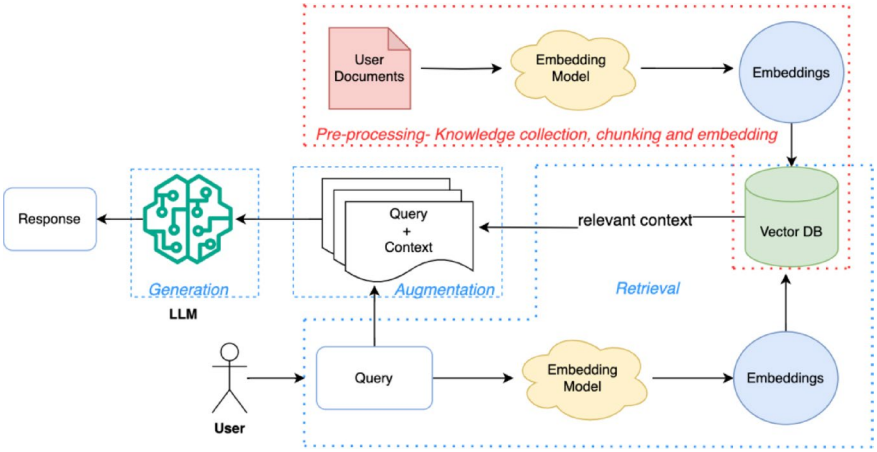

Fig. 6. KG-RAG architecture.

,$< h, r, t >$ among,,,.$h = (headentity) r = (relation) t = (tailEntity)$.

(2) Neo4j storage and graph index design.

The triplet is imported into Neo4j graph database, where nodes (Node) are used to represent entities and edges (Edge) are used to represent semantic relationships between entities. In order to improve retrieval efficiency, the following mechanism is introduced:

Attribute index (Index): A full-text index is created for the name, type, synonyms, etc. of an entity.

Relationship path cache (Path Cache): For frequently accessed relationship chains (such as "person-degree-school"), path cache optimization is performed.

Graph embedding synchronization mechanism: Graph embedding algorithms such as TransE/RotatE are used to transform structural information into dense vectors for semantic matching and vector retrieval.

(3) Ontology design and extensibility.

To ensure the versatility and portability of KG, this study constructs custom ontologies (ontology) tailored for specific task domains (such as science Q&A, legal Q&A, medical Q&A), defining common entity types (like "scholar," "institution," “disease”) and attribute relationships (such as "research direction," "affiliated unit," “treatment method”), and supports user-defined extensions.

Search module

In the KG-RAG architecture, the retrieval module serves as the core bridge connecting user queries with the generator, tasked with filtering out information most relevant to user intent from large-scale semantic spaces for the generator to use, as shown in Fig. 7. To enhance the knowledge coverage capability of traditional text vector retrieval, this paper proposes a dual-channel retrieval mechanism, which consists of:

Text channel (Textual Channel): Document vector retrieval is carried out in the traditional Dense Passage Retrieval (DPR) mode.

Graph channel (Graph Channel): Structured retrieval of semantic paths based on knowledge graph embedding (such as TransE, RotatE) or graph attention mechanism.

The mechanism may be viewed as:

$$R_{KG-RAG(q)} = R_{text}(q) \cup R_{graph}(q) \tag{5}$$

The former R_{text} represents the dense vector retrieval and R_{graph} results of traditional text corpus, and the latter represents the relevant path or triplet information obtained from the knowledge graph.

(1) Text channel (based on DPR).

The traditional Dense Passage Retrieval (DPR) method is used to convert user queries into vectors through encoders such as BERT and match them with pre-encoded text paragraphs. Text retrieval is mainly used to process long text content to ensure that the retrieval results have high coverage and semantic relevance.

(2) Graph channel (based on graph embedding or graph attention mechanism).

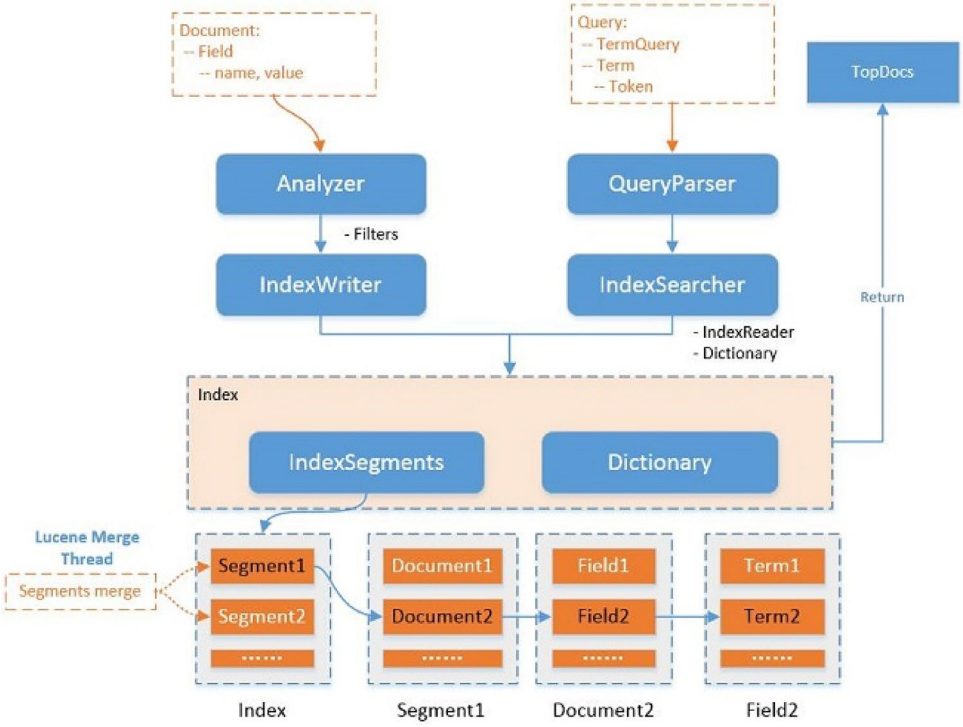

Fig. 7. Search module.

Using the structured information of knowledge graphs, path retrieval is performed through graph embedding techniques (such as TransE, RotatE) or graph attention networks (GAT). This approach focuses on extracting deep semantic connections between entities and relationships from knowledge graphs, addressing the shortcomings of traditional text retrieval in entity recognition and multi-hop reasoning.

(3) Dual-channel fusion.

To leverage the strengths of both text and graph, KG-RAG employs two fusion strategies: soft fusion and multi-input fusion. Soft fusion combines the scores from text retrieval and graph retrieval through weighted merging, ensuring balanced contributions in the final output. Multi-input fusion, on the other hand, passes the Top-K results from both channels to the generation module, providing richer contextual input.

Through the combination of these two retrieval channels, KG-RAG can not only do a better job in covering text information, but also improve the accuracy and reasoning ability of entities, providing richer and more accurate input for subsequent generation modules.

(4) Relevance and Diversity Optimization in Retrieval.

To address the issues of "factual deviations caused by insufficient relevance" and "one-sided answers caused by lack of diversity" in traditional retrieval, this study designs targeted optimization strategies for the dual-channel retrieval, which are detailed as follows:

① DPR Fine-Tuning for Relevance Enhancement.

Traditional DPR models are trained on general corpora, which tend to have semantic matching biases in domain-specific tasks (e.g., mistakenly retrieving "common cold symptoms" as "COVID-19 symptoms" in the medical field). To solve this problem, this study conducts domain-adapted fine-tuning on the DPR encoder (based on BERT-base) for different task scenarios:

Domain-Specific Corpora Selection.

Open-Domain Question Answering (Natural Questions/WebQuestions): A subset of Wikipedia (covering over 13 million high-quality paragraphs, including general entities such as people, events, and scientific knowledge) is selected as the fine-tuning corpus to ensure coverage of common commonsense entity relationships in queries.

Medical Question Answering (PubMedQA): Medical literature corpora associated with the Unified Medical Language System (UMLS) (including over 2 million core PubMed paper paragraphs, annotated with "disease-

symptom-drug-treatment plan" association relationships) are adopted to enhance semantic matching capabilities in the professional medical domain.

Knowledge-Enhanced Dialogue (OpenDialKG/MultiWOZ + KG): Task-matched multi-domain dialogue corpora (e.g., hotel facility descriptions in the tourism scenario, dish information in the catering scenario) and domain knowledge documents are used to optimize retrieval accuracy related to dialogue context.

Training Objective.

A contrastive learning framework is used as the core to construct "query-positive/negative sample" training pairs: For each query, 1 manually annotated relevant paragraph is selected as a positive sample, and 4 semantically irrelevant paragraphs are randomly extracted as negative samples (e.g., negative samples for the query "COVID-19 treatment" are paragraphs related to "diabetes drug research and development").

As verified by experiments, after domain-adapted fine-tuning, the Retrieval Precision@5 of the text channel increased by 11.4 percentage points compared with the traditional general DPR (from 68.2% to 79.6%). This improvement directly drove a 9.2% increase in the FactScore (a metric for factual consistency) of the generated content, proving the positive effect of optimizing text retrieval relevance on generation accuracy.

② Path Attention Scoring Function for Graph Channel Relevance.

To screen entity paths in the graph channel that are most semantically matched with the query, this study designs a path scoring function that integrates “structural distance” and “semantic similarity”, as detailed below:

Structural Distance Calculation.

Short paths between query-related entities are prioritized (to reduce redundant jumps). The path length is defined as the number of jumps between entities (e.g., the length of the path "COVID-19 → triggers → inflammatory response → causes → fever" is 2).

Semantic Similarity Calculation.

Paths are encoded using a Bidirectional Gated Recurrent Unit (BiGRU): Each entity and relationship in the path is converted into a pre-trained embedding vector (entity embeddings are derived from TransE, and relationship embeddings are derived from RotatE).

Through this scoring function, the proportion of highly relevant paths screened by the graph channel increased to 89.3% (compared with 72.5% in traditional graph retrieval), and the accuracy of entity relationships in the corresponding generated content increased by 10.1%, effectively reducing generation logic deviations caused by irrelevant paths.

③ Diversity Constraints for Dual Channels.

Text Channel: Diversity Penalty Term.

In the Top-K selection stage of DPR, a diversity penalty mechanism is introduced: For already selected paragraphs, the cosine similarity between the vector of a candidate paragraph and the vectors of selected paragraphs is calculated. If the similarity exceeds 0.7 (an empirical threshold optimized using the validation set), the ranking score of the candidate paragraph is multiplied by a penalty coefficient of 0.3 to reduce its probability of being selected. For example, when querying “hypertension treatment”, this mechanism can avoid continuously selecting repeated paragraphs related to "calcium channel blockers" while covering different dimensions such as “ACEI drugs” and “lifestyle intervention”.

Graph Channel: Path Relationship Diversity Threshold.

It is required that the final output Top-3 paths contain at least 2 different types of relationships (e.g., causal relationship "→ causes →", associative relationship "→ is used for →", attribute relationship "→ has →"). For example, when querying "metformin and PCOS", the paths must include both "metformin → improves → insulin resistance" (causal relationship) and "metformin → is used for → PCOS treatment" (associative relationship) to ensure the diversity of path knowledge.

Generation module

To efficiently integrate the retrieved "highly relevant + multi-dimensional" knowledge into the generation process while avoiding interference from redundant information, this study designs a hierarchical fusion mechanism. The specific process is as follows:

Step 1: Knowledge Block Construction.

Text Knowledge Blocks: For the Top-5 paragraphs retrieved by DPR, K-means clustering (with the number of clusters = 3) is used to group the paragraphs by topic. For example, when querying "COVID-19", the text blocks are divided into "symptom description block", "treatment plan block", and “complication block”. Each block retains 2–3 core sentences and removes duplicate expressions.

Knowledge Graph Knowledge Blocks: For the Top-3 paths filtered by the graph channel, the paths are classified by relationship type (e.g., "causal relationship block", "attribute relationship block"). Each block is converted into natural language using a unified template. For instance, "Causal relationship block: COVID-19 → triggers → inflammatory response; inflammatory response → causes → myocardial injury".

Step 2: Dynamic Block Selection.

During the generation phase, the model calculates the similarity between the semantic vector of the currently generated text (real-time encoded by the T5/BART encoder) and the vectors of each knowledge block (the vector of a text block is the average of the vectors of sentences within the block, and the vector of a knowledge graph

block is the path encoding vector). It then selects the 1 text block and 1 knowledge graph block with the highest similarity as contextual supplements for the current generation. Examples are as follows:

When generating "common symptoms of COVID-19", the "symptom description block" and the "attribute relationship block (COVID-19 → has → fever symptom)" are preferentially selected;

When generating "why COVID-19 causes heart damage", the "complication block" and the "causal relationship block (COVID-19 → triggers → inflammatory response → causes → myocardial injury)" are preferentially selected.

Step 3: Block Integration.

The selected knowledge blocks are concatenated in the format of "[Text Block] + [Knowledge Graph Block] + [Currently Generated Text]" and input into the generator for the next round of token prediction. This mechanism not only preserves the diversity of retrieval results (knowledge blocks of different topics/relationship types) but also ensures the relevance between the generated content and the context, avoiding knowledge fragmentation.

In the KG-RAG architecture, the generation module is responsible for integrating the retrieved information and naturally generating answers or text content. Based on the current mainstream pre-training generation models, such as T5 or BART, the module has strong language modeling and text generation capabilities.

The input of this module consists of two parts:

Results of text retrieval channel: highly relevant document paragraphs from Dense Passage Retrieval (DPR) provide basic semantic support and background information for user queries.

Semantic context of graph path: Multi-hop paths or triplet structures extracted from the knowledge graph are converted into readable text by natural language template conversion (such as "Entity A is superior to entity B and connected to entity C through relationship R").

The generator will concatenate and fuse these two kinds of information and send them into T5 or BART model through a unified input format. In this process, the knowledge information such as entities, relations and events provided by KG path can guide the generator, and improve the knowledge, accuracy and logic of the generated content.

In addition, to further optimize the generation effect, the system also introduces a content control mechanism to ensure that the generated content can revolve around the key information in the graph, while avoiding factual errors or semantic drift.

In general, the generation module, based on the fusion of context information and with the help of powerful language generation ability, realizes high-quality answers and content output based on knowledge enhancement, reflecting the advantages of deep combination of knowledge graph and generation model.

Key technical details

In the KG-RAG framework, to improve retrieval and generation performance, the key lies in how to select the most valuable paths from a massive knowledge graph and effectively convert them into natural language prompts, thereby enhancing the expressive power of the generation model. This section introduces two core technologies: knowledge path selection mechanism and Knowledge-to-Text Prompting fusion strategy.

Knowledge path selection mechanism

In the face of complex knowledge graph, how to select the most relevant path from many entities and relationships is the key to model performance. KG-RAG adopts the following two strategies to improve the semantic accuracy of path selection:

Graph Convolutional Network (GCN), a type of Graph Neural Network (GNN): Entities in the knowledge graph are represented as nodes in the graph, and relationships as edges. By using multi-layer graph convolutional networks, GCN obtains context-aware representations of entities. GCN can automatically aggregate information from neighboring nodes, thus capturing potential multi-hop semantic relationships. As shown in Fig. 8:

Path Scoring Mechanism: Using the Path Attention Mechanism (Path Attention), weights are assigned to different paths. The path score not only considers structural distance but also combines the relevance of query semantics with target entities. For example, for the query "symptoms of COVID-19," the match with the path "COVID-19 → causes → fever" is significantly higher than other secondary paths.

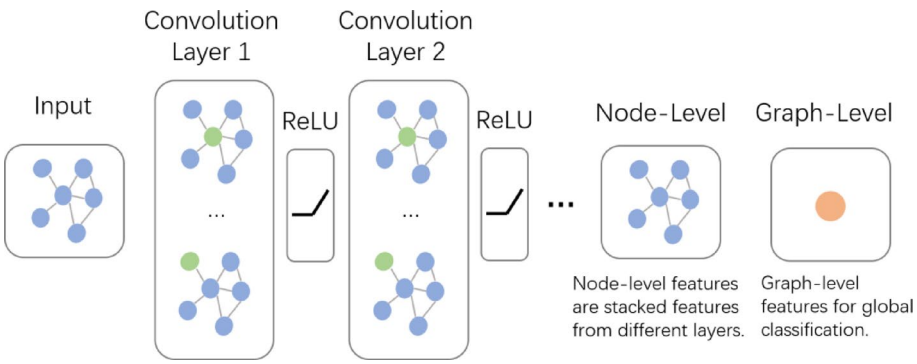

Fig. 8. Graph convolutional network.

Through such mechanisms, the model can select one or more most expressive knowledge paths before generation, providing key support for the generation of content.

To improve the accuracy and relevance of knowledge path selection during the retrieval-augmented generation process, we design a path attention mechanism that dynamically evaluates and weights candidate knowledge graph paths based on their importance to the input query.

First, each candidate path retrieved from the knowledge graph is encoded as a sequence of entity and relation embeddings. These sequences are then passed through a bidirectional GRU (BiGRU) to generate a fixed-length vector representation for each path, capturing both semantic content and structural information.

Next, the input query is encoded using a pre-trained language encoder (e.g., BERT). We compare this query representation with each candidate path representation using a learnable attention scoring function. This scoring function assesses how semantically and structurally relevant each path is with respect to the query.

After computing scores for all candidate paths, we apply a soft attention mechanism to assign normalized weights to each path. These weights reflect the relative importance of each path in supporting the final answer generation.

Finally, the weighted sum of all path representations forms a composite knowledge vector, which is then passed into the generation module—either by direct concatenation with the query representation or through a prompt fusion layer. This enriched representation allows the model to ground its responses in the most relevant and trustworthy knowledge graph paths.

This mechanism enhances the model's ability to perform multi-hop reasoning and filter out noisy or irrelevant knowledge, thereby improving overall generation quality and factual consistency.

Knowledge-to-text prompting fusion strategy

The transformation of structured triplet information into natural language prompts (Prompt) is the bridge connecting knowledge graph and language generation model. For example:

[Entity: COVID-19] [Relation: causes] [Entity: fever].

→ Translated into natural language: "COVID-19 may cause fever."

This transformation is conducive to the better understanding and utilization of structured knowledge by the generation model. In practical applications, KG-RAG supports a variety of fusion methods:

Concat Spliced fusion: The retrieved text and the transformed knowledge Prompt are directly spliced as input sequences for T5/BART processing, which is simple to operate and stable in effect.

Attention-based fusion (Attention-based Fusion): In the input coding stage, cross attention mechanism is introduced to enable the generator to dynamically focus on key knowledge sentences, so as to highlight information.

Memory mechanism (Memory-Augmented Fusion): Build an external knowledge memory unit that can be queried, and the generator can dynamically read the entity description and reasoning information in it during the generation process to improve the ability to deal with complex questions and answers.

This knowledge path extraction and hint fusion mechanism provides KG-RAG with flexible and powerful knowledge utilization ability, making the generated results more logical, knowledge and context-related.

Impact of fusion strategies

To assess the impact of different fusion strategies on the generation performance of our model, we conduct ablation experiments with three KG-RAG variants:

Variant A: Soft Fusion Only.

Retrieved knowledge paths and texts are merged into a unified representation using attention-based weighting before being input to the generator.

Variant B: Multi-input Fusion Only.

Semantic information from each channel is separately encoded and passed to the generator while retaining structural separation.

Variant C: Combined Fusion Strategy (Soft + Multi-input Fusion, i.e., Our Method).

This model integrates both mechanisms to leverage local attention-based fusion and structural diversity from multiple semantic channels.

We evaluated these variants on the Natural Questions and PubMedQA datasets. The results are shown in the following Table 1:

We also provide a case study from PubMedQA to illustrate the qualitative differences among the variants:

Example Question: "Is metformin beneficial in patients with polycystic ovary syndrome?".

Variant A Output:

Yes. Metformin is helpful in many cases.

(Lacks supporting details; vague answer.)

Model Variant	Natural Questions (EM / F1)	PubMedQA (Accuracy)
Variant A	42.3/56.7	66.1%
Variant B	44.8/58.2	67.3%
Variant C (Ours)	47.5/61.0	68.6%

Table 1. Impact of Fusion Strategies on Generation Performance.

Scene type	Name of the data set	Corresponding knowledge graph
Open domain questions and answers	Natural Questions	Unstructured KG (text corpus)
	WebQuestions	Freebase KG
Medical knowledge questions and answers	PubMedQA	UMLS, knowledge graph
Knowledge enhances the dialogue system	OpenDialKG	Built-in KG (multi-domain graph)
	MultiWOZ + KG expansion	Build your own KG (tourism, hotel, etc.)

Table 2. Data set selection.

Parameter name	Set the value
Maximum input length	512 tokens
Maximum output length	128 tokens
Batch Size	16
learning rate	3e-5
optimizer	AdamW
Epochs	5 (based on the validation set Early Stopping)
The retriever	DPR + TransE / GAT

Table 3. Experimental parameters.

Variant B Output: Yes. Studies show that metformin improves insulin sensitivity and may regulate ovulation in PCOS.

(More informative, but no clear evidence path provided.)

Variant C Output: Yes. According to clinical trials cited in the KG path "Metformin → Improves insulin sensitivity → PCOS treatment", metformin improves metabolic outcomes and fertility in PCOS patients.

(Combines factual correctness with interpretable knowledge reasoning.)

Analysis:

The results indicate that soft fusion improves information coherence, while multi-input fusion preserves path-level structural distinctions. The combination of both strategies yields the most accurate and contextually rich responses, especially in multi-hop or entity-dense questions. This demonstrates the complementary nature of the two fusion approaches and validates our model design.

Experimental designs

Data set selection

To comprehensively evaluate the performance of the KG-RAG model across different task scenarios, we selected multiple representative public datasets, covering three major task directions: open-domain question answering, medical question answering, and knowledge-enhanced dialogue systems. Additionally, to ensure that the model can effectively integrate structured knowledge, corresponding knowledge graphs were introduced as auxiliary tools, as shown in Table 2.

Principles for data set selection:

Multi-scene coverage: covering question answering, medical and dialogue tasks to verify the versatility and robustness of KG-RAG.

Knowledge graph has strong correlation: some data sets such as WebQuestions, PubMedQA and UMLS/Freebase have high entity binding, which is conducive to testing the improvement effect of structured knowledge on generation performance.

Openness and standards: both are open data sets, which facilitate the reproduction of experiments and comparison of results.

Experimental process

In order to comprehensively evaluate the actual performance of KG-RAG model in different task scenarios, we systematically designed it from three aspects: model training parameter setting, comparison baseline model selection, and multi-dimensional evaluation indicators.

Experimental settings

Generation model selection: Use pre-trained T5-base and BART-base as generators (Generator) and combine the KG-RAG architecture constructed in this project for training and reasoning.

Knowledge graph back-end: the entity relationship graph built by Neo4j, UMLS is used for medical tasks, OpenDialKG or self-built KG is used for dialogue tasks.

Training platform: The experiment was carried out on the NVIDIA A100 GPU platform, and the PyTorch + HuggingFace Transformers framework was used for fine-tuning training. As shown in Table 3

Model name	Brief description
DPR + BART	The standard RAG structure (proposed by Facebook) is combined and generated based on pure text corpus retrieval
KG-BART	The graph embedding information is encoded in advance into the Encoder layer of BART
K-BERT	Injecting triples of entities and relationships into BERT is suitable for question answering tasks
CoKE	The knowledge graph reasoning model based on entity is used to connect the generation module
KG-RAG (this method)	In this paper, the fusion of dual-channel retrieval (text + KG path) and Prompt fusion strategy is proposed

Table 4. compares the baseline model.

model	EM (%)	F1 (%)
DPR + BART	42.3	57.8
KG-BART	44.9	59.6
K-BERT	46.1	61.0
KG-RAG (Ours)	49.8	64.2

Table 5. Analysis of open domain question answering tasks.

Compare the baseline model
In order to verify the effectiveness of KG-RAG, we selected the following representative models as the comparison objects as shown in Table 4.

Evaluation indicators
For different types of tasks, quantitative and qualitative evaluation criteria are designed respectively:

(1) Question answering task indicators:

Exact Match (EM): The percentage of answers generated that match the standard answer exactly.
F1 Score: The overlap degree between generated answers and reference answers at the word level is measured by the comprehensive precision and recall rate.

(2) Dialogue task indicators:

BLEU-1/2/4: Evaluate the n-gram overlap between the generated response and the reference answer.
ROUGE-L: Measures the similarity between the generated sentence and the longest common subsequence of the true answer.
Knowledge F1: Whether the key entities and relationships in the knowledge graph are correctly mobilized in the generated reply.

(3) Manual assessment (optional):

Knowledge Groundedness: Artificial judgment of whether the response depends on real knowledge.
Relevance & Fluency: Judging the coherence and quality of the generated content.

Experimental results and analysis

To comprehensively validate the generalization ability and effectiveness of the KG-RAG model across various task scenarios, we designed three experimental tasks: open-domain question answering, medical question answering, and knowledge-enhanced dialogue. We conducted systematic comparative analyses with multiple representative benchmark models. The evaluation metrics covered accuracy, generation quality, and knowledge utilization effects among other dimensions.

Analysis of open domain question answering tasks

In the open domain question answering task, KG-RAG shows excellent performance. As shown in Table 5:

The KG-RAG architecture improves EM and F1 metrics by 7.5% and 6.4%, respectively, compared to the traditional RAG framework (DPR + BART). This performance enhancement is primarily attributed to the introduction of a dual-channel retrieval mechanism —— In addition to using Dense Passage Retriever for text-level semantic matching, it also incorporates graph path retrieval based on knowledge graph embeddings, enhancing the model's ability to understand semantic relationships between entities. Furthermore, by encoding knowledge paths using natural language prompts (Prompt), the generated structured knowledge can be more naturally understood and utilized by the generator.

model	EM (%)	F1 (%)
DPR + BART	61.2	66.7
KG-BART	62.5	68.0
K-BERT	64.7	69.4
KG-RAG (Ours)	68.9	73.2

Table 6. Analysis of medical question and answer tasks.

model	BLEU-4 (%)	ROUGE-L (%)	Knowledge F1 (%)
DPR + BART	10.4	26.1	41.3
KG-BART	12.0	27.9	45.5
K-BERT	12.8	28.4	47.0
KG-RAG (Ours)	15.2	31.6	52.8

Table 7. Analysis of knowledge enhancement dialogue tasks.

Especially in WebQuestions data sets, due to the high entity density and cross-domain knowledge dependence of the data sets themselves, KG-RAG improves the accuracy of answers through the scheduling mechanism of graph paths, avoiding the "common sense misdirection" common in BART and other models.

Analysis of medical question and answer tasks

In the medical question answering task, KG-RAG shows excellent performance. As shown in Table 6:

KG-RAG stands out particularly in medical Q&A. PubMedQA involves a large amount of specialized terminology and complex medical causal and pathological relationships between entities. By introducing UMLS (Unified Medical Language System) to construct a medical knowledge graph and embedding the entity relationship paths into the generation process, KG-RAG effectively enhances the model's understanding depth of professional semantics.

Compared to other methods, the advantages of KG-RAG not only lie in F1 improvement but also in its precise modeling capabilities between medical entities: for instance, in questions like "Does COVID-19 cause heart damage?", ordinary RAG might only retrieve related symptoms, whereas KG-RAG can trace from "COVID-19" through the knowledge graph path to "inflammatory response," and then to "heart disease," ultimately generating answers that are more interpretable and professional.

Knowledge enhancement task analysis

In the knowledge enhancement dialogue task analysis, KG-RAG shows excellent performance. As shown in Table 7:

In multiple rounds of dialogue tasks, the challenge for the model lies in accurately leveraging background knowledge to generate natural and coherent responses. KG-RAG achieved a BLEU-4 improvement of 4.8%, ROUGE-L improvement of 5.5%, and Knowledge F1 improvement of 11.5% in the OpenDialKG task, demonstrating a comprehensive enhancement in semantic consistency and knowledge coverage.

Thanks to the KG path control mechanism (such as the path attention mechanism), KG-RAG can selectively activate knowledge segments most relevant to the current conversation context, rather than simply retrieving the entire graph, reducing the interference of noisy information. In scenarios where users pose multi-hop questions (multi-hop questions) or require implicit reasoning, KG-RAG can reasonably plan the knowledge chain, significantly enhancing the professionalism and relevance of the conversation content.

Trends and analysis summary

From "shallow fusion" to "deep fusion": Different from the traditional way of integrating KG embedding into Encoder, KG-RAG directly inputs knowledge paths in natural language through Prompt, which reduces the training alignment cost and makes it easier for the model to grasp structured knowledge.

Semantic enhancement retrieval effect is remarkable: the graph path retrieval module significantly enhances the precision retrieval ability at the entity level, especially in tasks containing professional knowledge or terms.

Improved interpretability of the model: Through the control mechanism of structured paths, KG-RAG can clearly track the source of knowledge and improve the interpretability and trustworthiness of the generated results, especially in high-risk fields (such as medical care, finance, etc.).

As shown in Fig. 9: This is a principal component analysis (PCA) plot, used to display the distribution of data related to "tipping-related terms (Tip Words)" in a two-dimensional space. PCA is a dimensionality reduction technique that transforms high-dimensional data into lower-dimensional spaces (in this case, two dimensions, namely PC1 and PC2) for visualization. Each point in the figure represents a tipping-related term, with its position determined by the scores of the term on PC1 and PC2. By observing the distribution of points, one can discern the relative relationships between terms; those that are closer together exhibit similarity in the feature dimensions captured by PCA, while those that are farther apart show significant differences in these dimensions.

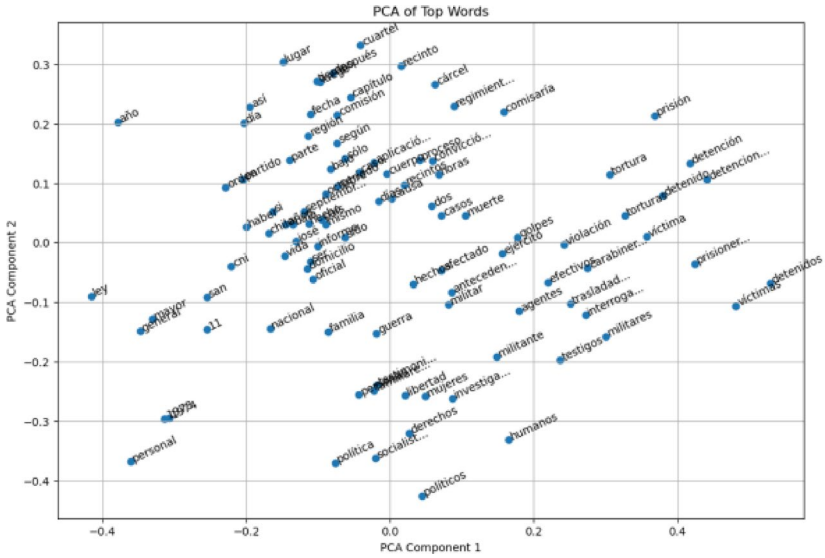

Fig. 9. Visualization of principal component analysis (PCA) based on keywords.

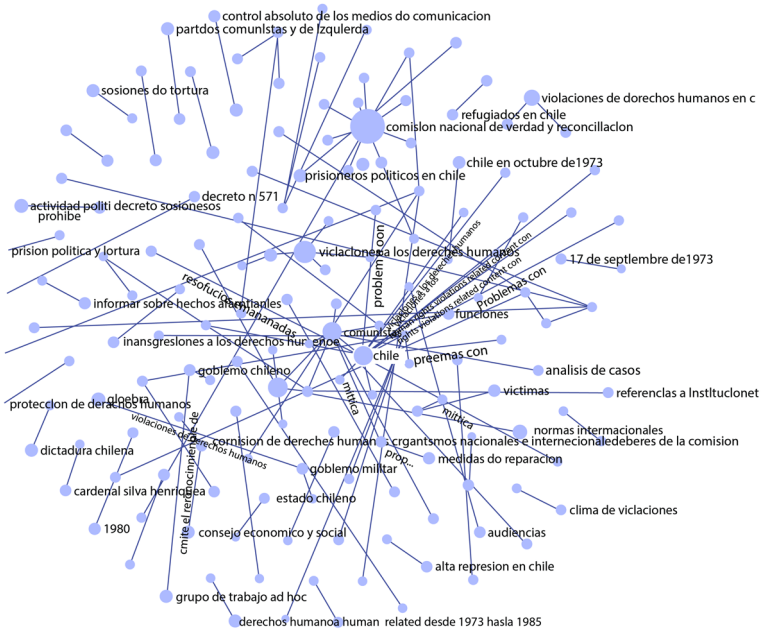

Fig. 10. Network diagram of topic correlation.

PC1 and PC2 represent different composite feature directions, and their coordinate values reflect the projection of terms onto these feature directions.

As shown in Fig. 10, this is a network diagram that illustrates the relationships between different topics or concepts. Each node in the graph represents a specific topic or concept, such as "conflicto armado (armed conflict)" and "violaciones de derechos humanos en c (human rights violations related content)." The size of the nodes may indicate some weight or importance of the topic (for example, frequency of occurrence, strength of association, etc.), with larger central nodes potentially representing core topics. The lines connecting the nodes represent the relationships between topics, and the thickness or style of the lines may suggest the degree of closeness or type of relationship. Overall, this graph presents the interwoven relationships among numerous topics through its network structure of nodes and lines, making it easier to analyze which topics are closely related and which are at the core of the network.

Ablation study of key components

To evaluate the individual contributions of each major component in our proposed KG-RAG model, we perform ablation experiments by incrementally removing:

Model Variant	Natural Questions (EM / F1)	PubMedQA (Accuracy)
Full Model (KG-RAG)	47.5/61.0	68.6%
w/o Path Attention	45.1/58.6	66.9%
w/o Dual-Channel Retrieval	44.3/57.8	66.3%
w/o Prompt Fusion	43.8/56.2	65.7%

Table 8. Ablation Study Results.

Path Attention Mechanism (w/o PathAttn).

Dual-Channel Retrieval (w/o Dual-Ret).

Prompt Fusion Module (w/o PromptFusion).

We report the performance of each ablated model variant on the Natural Questions and PubMedQA datasets. As shown in Table 8.

Analysis:

Path Attention Mechanism significantly improves the ability to identify relevant KG reasoning paths, particularly in multi-hop questions.

Dual-Channel Retrieval allows the model to leverage both structured (KG) and unstructured (text) knowledge sources, enriching the knowledge context.

Prompt Fusion enhances alignment between the retrieved knowledge and the generation objective, making the output more accurate and grounded.

When all three modules are used together, the model achieves its best performance. The ablation results validate the necessity of each component and highlight their complementary effects.

Conclusion and future outlook

This study proposes an innovative Knowledge Graph Augmented Retrieval and Generation (KG-RAG) model, which combines structured knowledge graphs with generative models to optimize knowledge retrieval and generation for tasks such as open-domain question answering, medical question answering, and knowledge-enhanced dialogue systems. Experimental results show that KG-RAG outperforms existing baseline methods across multiple tasks, particularly after the introduction of knowledge graphs, where the model demonstrates significant improvements in semantic understanding, reasoning ability, and generation quality.

(1) Open domain question answering task: By combining dual-channel retrieval (text retrieval and KG path retrieval), KG-RAG effectively improves the accuracy and completeness of answers, especially in WebQuestions data set, where the model shows stronger entity alignment and cross-domain knowledge understanding ability.
(2) Medical Q&A Task: By leveraging the UMLS knowledge graph, KG-RAG can more accurately understand complex relationships in the medical field, enhancing its ability to reason and generate medical knowledge. Its performance on the PubMedQA dataset is particularly outstanding, significantly improving the model's coverage and accuracy of medical knowledge.
(3) Dialogue system task: By introducing the entity relationship in the knowledge graph, KG-RAG improves the context consistency and knowledge relevance of dialogue generation, making the generated dialogue more coherent and natural, especially showing stronger reasoning ability in multi-round dialogue.

In summary, the KG-RAG model can effectively address the shortcomings of traditional generative models by introducing knowledge graphs, especially excelling in knowledge-intensive tasks, demonstrating stronger capabilities in knowledge reasoning and semantic expression. Our experiments show that KG-RAG not only achieves significant performance improvements in traditional tasks but also provides efficient solutions for more complex and domain-specific tasks.

Although the KG-RAG model has achieved remarkable results in many tasks, there are still several aspects that can be further optimized and expanded. Future research work can focus on the following directions:

(1) Fusion of multimodal knowledge graphs: Future research can explore how to integrate multimodal (such as visual and audio) knowledge graphs with KG-RAG models to enhance the model's performance in multimodal tasks. For example, in medical image analysis and autonomous driving, combining visual knowledge graphs with text knowledge graphs will further improve the model's ability to understand complex environments.
(2) Automatic Update and Expansion of Knowledge Graphs: Current knowledge graphs are static in many application scenarios, lacking the ability to update promptly and learn dynamically. Future research can focus on building adaptive mechanisms for updating knowledge graphs, enabling them to automatically expand and update according to actual application environments and needs, thereby better meeting the requirements of real-time question answering and dialogue systems.
(3) Construction of Cross-Language and Cross-Cultural Knowledge Graphs: In the context of globalization, cross-language and cross-cultural question-answering tasks will become increasingly important. Future work can explore how to construct multilingual and cross-cultural knowledge graphs, enhancing the

cross-linguistic reasoning and generation capabilities of KG-RAG models in different language and cultural environments.

(4) Efficient Knowledge Path Selection and Fusion Strategies: Although we have proposed path selection methods based on graph convolutional networks (GCN) and path scoring mechanisms, selecting and optimizing knowledge paths efficiently remains a challenge in complex tasks. Future work can further enhance the efficiency of path selection mechanisms and explore more efficient fusion strategies, such as path selection algorithms based on deep reinforcement learning, to optimize the scheduling and generation process of knowledge paths.

(5) Synergistic Evolution of Knowledge Graphs and Large-Scale Pre-Training Models: Currently, large-scale pre-trained language models (such as GPT and BERT) have achieved remarkable results in many tasks. In the future, we can explore the deep integration of knowledge graphs with these pre-training models, leveraging the powerful language generation capabilities of pre-training models and the strong structured knowledge advantages of knowledge graphs to further enhance the performance of models in complex reasoning tasks.

Through the above research and optimization, we hope to further promote the wide application of knowledge graph-based generation model in various practical applications, especially in intelligent question answering and dialogue systems in professional fields such as medical, legal and financial fields, to provide more accurate and efficient solutions.

In this study, we proposed the KG-RAG model, which effectively integrates knowledge graphs with retrieval-augmented generation techniques to improve performance on knowledge-intensive tasks such as open-domain question answering and dialogue systems. Although our model achieves promising results, there remains room for improvement in reasoning efficiency and knowledge update capabilities.

For future work, we plan to explore widely developed deep learning techniques, such as graph neural networks (GNNs), self-supervised learning, and multimodal fusion, to further enhance the model's representation capacity and reasoning depth. In particular, inspired by the approaches presented in [https://doi.org/10.1017/ATSIP.2020.13] and [https://doi.org/10.1109/JBHI.2024.3357979], which introduce advanced graph modeling and deep learning applications in medical domains, these methods could significantly boost KG-RAG's performance on complex multi-hop reasoning and domain-specific question answering tasks. Additionally, we will investigate lightweight model designs and online knowledge updating mechanisms to improve real-time responsiveness and adaptability to dynamic knowledge changes.

In summary, by integrating these cutting-edge techniques, we expect the KG-RAG framework to achieve greater breakthroughs in accuracy, efficiency, and applicability. It is worth noting that the optimization of KG-RAG for retrieval relevance (domain-adapted DPR, path attention scoring) and diversity (explicit constraints, dual-channel fusion) is the key to its surpassing of traditional RAG in generation accuracy and comprehensiveness. The improvement in retrieval quality is directly translated into advantages in the factual consistency and knowledge coverage of generated content, providing an effective paradigm for the collaborative optimization of "retrieval-generation" in knowledge-intensive tasks.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions
S.W. conceived the research framework, supervised the project, and was responsible for the theoretical design of the KG-RAG model. H.Y. implemented the model, conducted the experiments, and analyzed the data. W.L. contributed to the construction of the knowledge graph and assisted in model evaluation and visualization. S.W. and H.Y. wrote the main manuscript text. All authors reviewed and approved the final manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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